

Attention Mechanisms in Sequence Models

Abstract

We introduce a sequence transduction architecture that dispenses with recurrence entirely and relies exclusively on a scaled dot-product attention operator. On the WMT 2014 English-to-German benchmark the model reaches 28.4 BLEU while training in a fraction of the time required by the recurrent baselines it replaces.

The central claim of this work is that alignment quality does not depend on sequential computation. Positional information enters the model through fixed sinusoidal encodings rather than through the order in which tokens are processed.

Method

Each encoder layer applies multi-head attention followed by a position-wise feedforward network. Residual connections wrap both sublayers and layer normalisation is applied to the sum.

The scaled dot-product operator divides the logits by the square root of the key dimension. Without this rescaling the softmax saturates for large key dimensions and the resulting gradients become vanishingly small, which is the specific failure this normalisation prevents.

We train with the Adam optimiser and a warmup schedule that increases the learning rate linearly for the first four thousand steps, then decays it proportionally to the inverse square root of the step number. Label smoothing of 0.1 is applied throughout.

Results and Ablations

Ablation group 1.

Removing the residual connection degrades convergence sharply.
We repeat this measurement across three independent seeds.
Halving the number of attention heads costs 0.9 BLEU on the test set.
We repeat this measurement across three independent seeds.
Sharing the embedding and output projection saves many parameters.
We repeat this measurement across three independent seeds.
Dropout on the attention weights is the strongest single regulariser.
We repeat this measurement across three independent seeds.
Increasing model depth beyond twelve layers yields no further gain.
We repeat this measurement across three independent seeds.
Beam search with a width of four outperforms greedy decoding.
We repeat this measurement across three independent seeds.

Ablation group 2.

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Ablation group 3.

Removing the residual connection degrades convergence sharply.
We repeat this measurement across three independent seeds.
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Beam search with a width of four outperforms greedy decoding.
We repeat this measurement across three independent seeds.

Ablation group 4.

Removing the residual connection degrades convergence sharply.
We repeat this measurement across three independent seeds.
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We repeat this measurement across three independent seeds.
Sharing the embedding and output projection saves many parameters.
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Dropout on the attention weights is the strongest single regulariser.
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We repeat this measurement across three independent seeds.
Beam search with a width of four outperforms greedy decoding.
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The complete hyperparameter sweep is reproducible from the seed table.